**Reader:** I see. If, for example, we wanted to pass from $y=f(x)$ to $y=f(\frac{1}{x})$, the function pictured in Fig. 11 would be transformed as follows:

$$\boxed{y=\frac{4/x^2-1}{1/|x|+1}}$$

**Author:** Correct. Now try to find $y=f(x)$ if $\boxed{2f(\frac{1}{x})-f(x)=3x}$ (Eqn 1)

**Reader:** I am at a loss.

**Author:** As a hint, I suggest replacing $x$ by $\frac{1}{x}$ (Sera)

**Son:** $2f(x)-f(\frac{1}{x})=\frac{3}{x}$ (Eqn 2)

$4f(x)-2f(\frac{1}{x})=\frac{6}{x}$ (Eqn 3).

Add Eqn (1) and (3): $3f(x)=3x+\frac{6}{x} \Rightarrow \boxed{f(x)=x+\frac{2}{x}}$

**Author:** Perfectly true.

**Reader:** In connection with your comment about the numerical function as a "black box" generating a numerical output in response to a numerical input, I would like to ask whether other types of "black boxes" are possible in calculus.

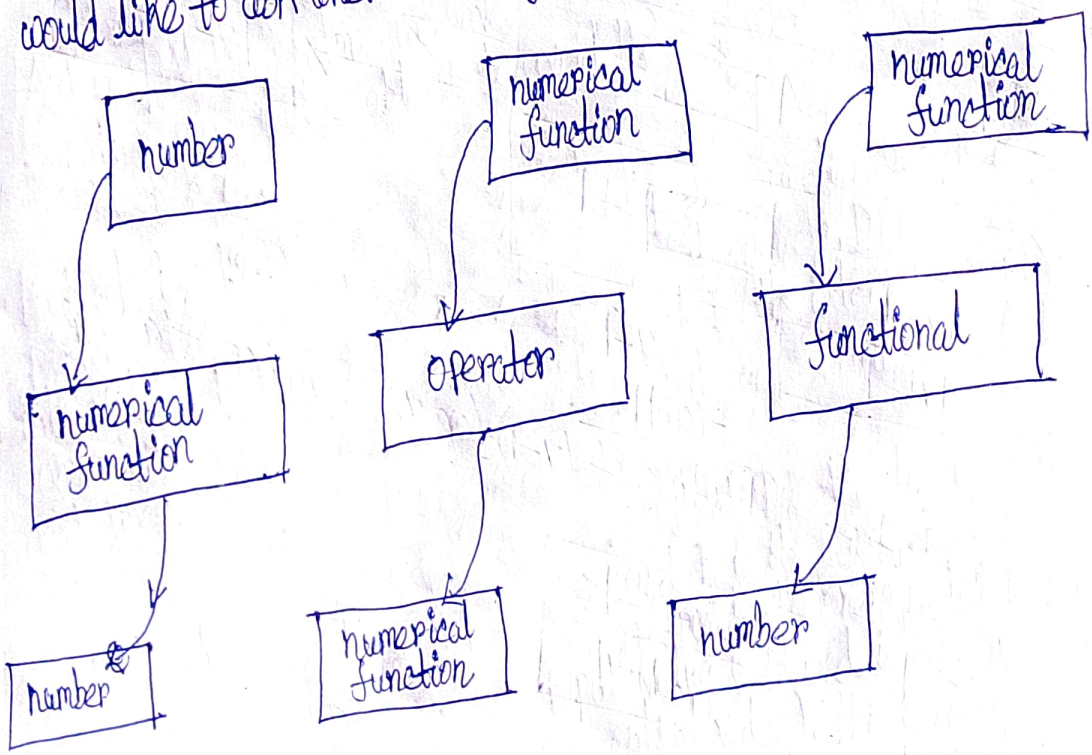


Fig. 12